

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	customer_id	age	gender	item_purchased	category	purchase_amount	location	size	color	season	review_rating	subscription_status	shipping_type	discount
▶	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes
	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes
	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes
	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes
	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes
	6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Standard	Yes
	7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Shipping	Yes
	8	27	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes	Free Shipping	Yes
	9	26	Male	Coat	Outerwear	97	West Virginia	L	Silver	Summer	2.6	Yes	Express	Yes
	10	57	Male	Handbag	Accessories	31	Missouri	M	Pink	Spring	4.8	Yes	2-Day Shipping	Yes
	11	53	Male	Shoes	Footwear	34	Arkansas	L	Purple	Fall	4.1	Yes	Store Pickup	Yes
	12	30	Male	Shorts	Clothing	68	Hawaii	S	Olive	Winter	4.9	Yes	Store Pickup	Yes
	13	61	Male	Coat	Outerwear	72	Delaware	M	Gold	Winter	4.5	Yes	Express	Yes
	14	65	Male	Dress	Clothing	51	New Hampshire	M	Violet	Spring	4.7	Yes	Express	Yes

previous_purchases	payment_method	frequency_of_purchases	age_group	purchase_frequency_days
14	Venmo	Fortnightly	Middle Aged	14
2	Cash	Fortnightly	Young Adult	14
23	Credit Card	Weekly	Middle Aged	7
49	PayPal	Weekly	Young Adult	7
31	PayPal	Annually	Middle Aged	365
14	Venmo	Weekly	Middle Aged	7
49	Cash	Quarterly	Senior	90
19	Credit Card	Weekly	Young Adult	7
8	Venmo	Annually	Young Adult	365
4	Cash	Quarterly	Middle Aged	90
26	Bank Transfer	Bi-Weekly	Middle Aged	NULL
10	Bank Transfer	Fortnightly	Young Adult	14
37	Venmo	Fortnightly	Senior	14
31	PayPal	Weekly	Senior	7

- **Missing Data Handling:** Checked for null values and imputed missing values in the `Review Rating` column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
 - Created **age_group** column by binning customer ages.
 - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender	revenue
▶	Male	157890
	Female	75191

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id	purchase_amount
▶	2	64
	3	73
	4	90
	7	85
	9	97
	12	68
	13	72
	16	81
	20	90
	22	62
	24	88
	29	94
	32	79
	33	67

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased	avg_product_rating
▶	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Skirt	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

Result Grid		Filter Rows:	Exp
	shipping_type	round(avg(purchase_amount),2)	
▶	Express	60.48	
	Standard	58.46	

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status	customer_count	avg_spend	total_revenue
▶	Yes	1053	59.49	62645
	No	2847	59.87	170436

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased	discount_rate
▶	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment	number of customers
▶	loyal	3116
	returning	701
	new	83

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

	item_rank	item_purchased	category	total_orders
▶	1	Jewelry	Accessories	171
	2	Sunglasses	Accessories	161
	3	Belt	Accessories	161
	1	Blouse	Clothing	171
	2	Pants	Clothing	171
	3	Shirt	Clothing	169
	1	Sandals	Footwear	160
	2	Shoes	Footwear	150
	3	Sneakers	Footwear	145
	1	Jacket	Outerwear	163
	2	Coat	Outerwear	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

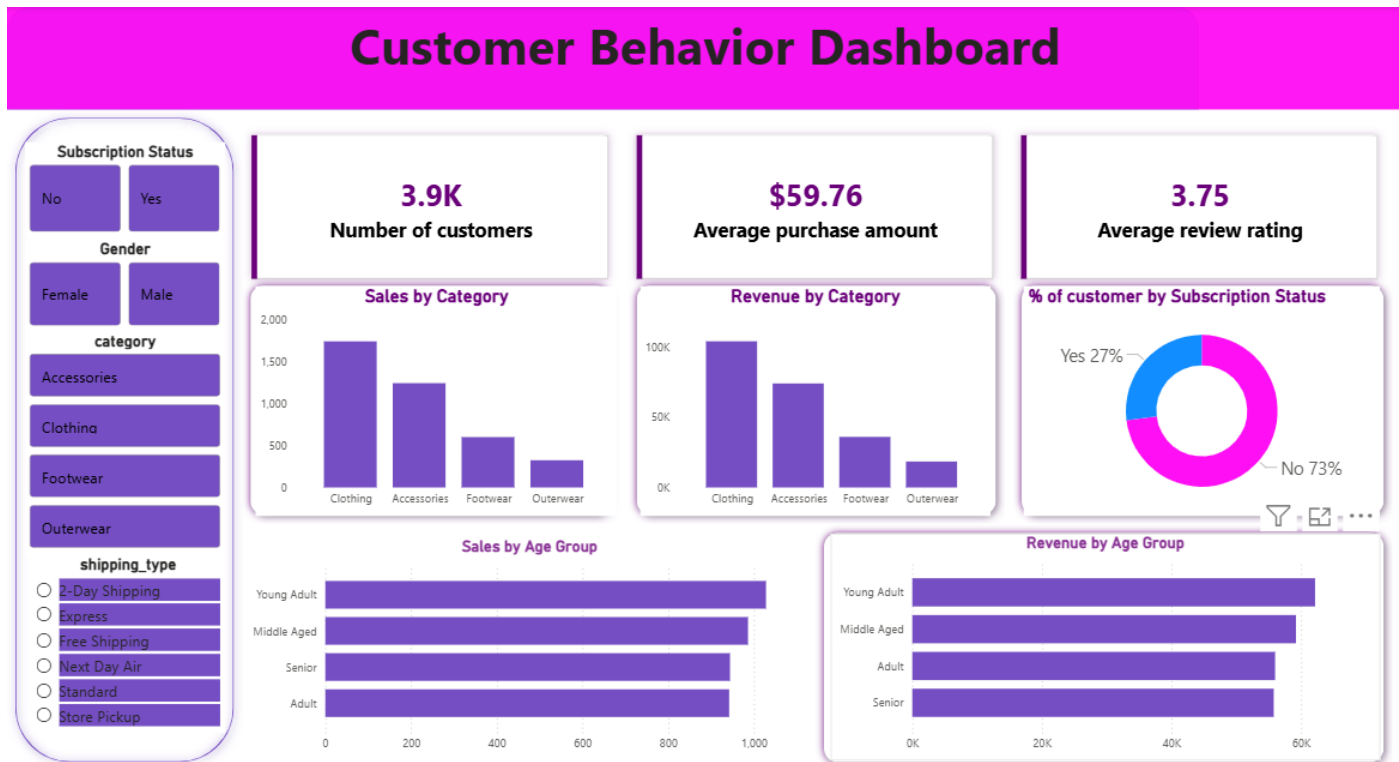
	subscription_status	repeat_buyers
▶	Yes	980
	No	2583

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_group	total_revenue
▶	Young Adult	62143
	Middle Aged	59197
	Adult	55978
	Senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.

- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.